

RESEARCH ARTICLE

Biopesticide Exposure Increases Aggression Without Impairing Nestmate Recognition in a Social Paper Wasp

Livia De Fazi^{1,2} | Rita Cervo¹ | David Baracchi¹  | Claudia Bruschini¹  | Federico Cappa¹ 

¹Department of Biology, University of Florence, Florence, Italy | ²Department of Agriculture, Food and Environment, University of Pisa, Pisa, Italy

Correspondence: Federico Cappa (federico.cappa@unifi.it) | Livia De Fazi (livia.defazi@phd.unipi.it)

Received: 15 January 2025 | **Revised:** 29 April 2025 | **Accepted:** 29 May 2025

Funding: This work was supported by the Italian Ministry of Universities and Research (MUR) (Protocol No 2022LJN3LE; Cup: B53D23012130006) and European Commission, project—PON Research and Innovation (2014–2020).

Keywords: aggressiveness | *Beauveria bassiana* | cuticular hydrocarbons | *Polistes dominula* | sublethal effects

ABSTRACT

Biopesticides have gained increased attention as sustainable substitutes for synthetic pest control products. Nonetheless, in recent years, a growing body of research has highlighted their undesired effects on non-target organisms, including sublethal impacts on social behavior. Here, we investigate the potential adverse effects of a common biopesticide, the entomopathogenic fungus *Beauveria bassiana*, on the nestmate recognition ability of the paper wasp *Polistes dominula*. Nestmate recognition, based on individual cuticular hydrocarbons (CHCs) profiles, plays a key role in the colony integrity of social insects, avoiding foreign intrusion and the spread of pathogens inside the nest. *P. dominula* workers were topically exposed to the biopesticide, and subsequent presentation behavioral assays and GC–MS chemical analyses were carried out to evaluate possible changes both in the nestmate recognition ability and chemical profile induced by *B. bassiana*. Biopesticide exposure altered the CHCs profile of exposed wasps and triggered increased aggression toward both exposed nestmates and foreign individuals, while not disrupting the colony nestmate recognition system. The higher aggressive response toward exposed wasps likely represents an adaptive strategy to prevent the entrance of pathogen-infected individuals into the nest. The present study demonstrates how an integrated approach based on behavioral bioassays and chemical analyses can effectively evaluate the sublethal effects of plant protection products on non-target organisms.

1 | Introduction

The ongoing anthropogenic environmental degradation and the resulting loss of biodiversity are one of the major issues of the Anthropocene, threatening the balance of planetary ecosystems and leading to a rapid and unprecedented global decline in biomass and genetic diversity [1, 2]. Insects face an alarming threat of extinction, with an estimated 40% of known species at risk. This decline is evident at local, regional, and global scales, affecting biomass, species richness, and abundance [3–5]. Many are the drivers of this “Insect Apocalypse” [6] and among them, the massive use of synthetic pesticides in agriculture poses an undeniably considerable threat to non-target beneficial insects [4].

To reduce the deleterious impact of synthetic pesticides, more sustainable alternatives, such as biopesticides, have been introduced into agriculture over the past decades and they are progressively replacing synthetic agrochemicals for crop protection [7, 8]. These natural control agents are considered to be eco-friendly and safe for humans and non-target organisms [9], helping to maintain ecosystems and biodiversity [10], and achieving efficacy comparable to synthetic substances in Integrated Pest and Pollinator Management (IPPM) strategies [11, 12]. However, although the impact of biopesticides on beneficial insect populations is often considered negligible [13], their application in the field can still have detrimental effects on non-target species [14–17].

Entomopathogenic fungi (EF) are among the most extensively used biopesticides [18, 19]. Among these, the fungus *Beauveria bassiana* has taken a pivotal role in the field due to its ability to selectively infect and kill multiple arthropod pests [19, 20]. First used in the 1960s in Colorado and Russia in the product Boverin [21], its use has nowadays spread worldwide, with numerous strains being used. In Europe alone, eight different strains have been approved from 2009 to 2022 [22]. Commercially available products often contain strains of *B. bassiana* selected to be specific to the target species and rarely pathogenic to non-target species [18, 23]. However, copious literature has highlighted the adverse effects that the fungus can have on some non-target species, in particular, pollinators (for reviews, see [14, 24]). Sublethal effects of *B. bassiana* so far documented on these taxa include alterations in gene expression, midgut histology, orientation capacity, learning and memory, foraging activity, chemical cuticular profile, and nestmate recognition abilities [14].

Recently, attention has begun to expand beyond pollinators to other beneficial insects that provide ecosystem-regulating services, such as social wasps [25]. Indeed, social wasps often nest and forage in agroecosystems, where they act as natural control agents, preying on a vast array of insect pests [25, 26], and during their foraging or nesting activities, they can be exposed to plant protection products [14, 27]. Pioneering studies have shown that biopesticides like the Neem tree-derived azadirachtin and *B. bassiana* can affect the survival, locomotor activity, reproduction, and social interactions of social wasps [27–29]. Yet, despite the increasing interest in these social Hymenoptera, scientific evidence regarding the real impacts that EF can have on physiological, functional, and behavioral traits, especially at the colony level, remains limited.

Social recognition, particularly, the ability to discriminate between nestmates and foreign individuals, is a crucial feature for the success of social insect colonies. This recognition is primarily driven by the perception of chemical cues, mainly cuticular hydrocarbons (CHCs), which play a critical role in acquiring information about conspecific individuals [30, 31]. An altered nestmate recognition system is likely to jeopardize colony integrity and survival. For example, failing to identify contaminated nestmates might enable these contaminants to infiltrate and spread throughout the colony [32]. Similarly, the impaired ability to detect foreign individuals could increase the entry of outsiders that might exploit colony resources [31, 33]. On the other hand, rejection of exposed nestmates may prevent pathogens or contaminants within the nest, but it could also reduce the available workforce, posing significant threats for species with colonies comprised of small numbers of workers [34].

Previous research has demonstrated that exposure to *B. bassiana* can induce alterations in CHCs, which in turn disrupt the colony recognition system. This disrupting effect has been observed in honey bees [33] and stingless bees [34]. Similar to many social insects, the paper wasp *Polistes dominula* primarily relies on chemical cues for various forms of social discrimination [35–37]. Due to their ability to detect subtle changes in the CHC profiles of conspecifics [37] and the small colony size, *P. dominula* can serve as a valuable model for testing the effects of

biopesticides on social recognition. Both foundresses and workers forage intensively for prey and nest-building materials, often visiting crop fields in search of insect pests (i.e., caterpillars) where they can be exposed to plant protection products [38]. Although *P. dominula* prefers to nest in sheltered locations, *Polistes* nests lack the protective envelope of other social wasps [39], making colonies built in the proximity of sprayed fields susceptible to pesticide exposure.

In this study, we investigate, for the first time, the potential impact of the biopesticide *B. bassiana* on the colony recognition system of the paper wasp *P. dominula*. Our aim is to assess not only whether the fungus alters the colony's acceptance threshold toward exposed individuals, but also to examine how the biopesticide affects the colony's recognition abilities.

2 | Materials and Methods

2.1 | *Polistes Dominula* Life Cycle

Polistes species are primitively eusocial paper wasps with a small colony and functional castes [40–42]. The species *P. dominula* is common and widespread in temperate regions where it can be an effective biocontrol agent due to its generalist predatory habits [25, 43, 44]. Additionally, it represents a promising candidate as a bioindicator of environmental quality and pollution [45]. In the annual colony cycle, reproductive females (i.e., foundresses) leave their overwintering shelters in early spring to establish new nests. The preemergence phase of the colony ends with the emergence of workers, usually in late spring. Then, several generations of adult workers emerge and remain active on the nests until late summer, taking over the tasks of nest building, provisioning, and defense. Reproductive individuals (i.e., sexuals) emerge in late summer, with mating occurring in autumn. Mated females then enter winter diapause, and the cycle starts anew the following spring [41].

2.2 | Colonies Collection and Laboratory Rearing

Colonies of *P. dominula* ($N=49$) in the preemergence phase (i.e., before the emergence of the first generation of workers) were collected in May 2022 from five different localities surrounding Florence, Italy. Colonies were then transferred to the laboratory and individually attached to the ceiling of $15\times 15\times 15$ cm glass boxes and maintained at $25^{\circ}\text{C}\pm 2^{\circ}\text{C}$ under an artificial timed light–dark cycle (L:D=12:12). Each colony box was supplied with *ad libitum* sugar, larvae of the flesh fly *Sarcophaga carnaria*, and water, and provided with blotting paper for nest construction. Foundresses were marked on their wings with acrylic colors (Humbrol Enamel) to distinguish them from newly emerged workers that were marked upon emergence with different colors on their wings to assess their age. All workers used in recognition bioassays and chemical analyses were at least 7 days old to ensure the full development of their chemical profile [46]. All colonies tested in recognition bioassays were in the post-emergence phase with a comparable number of workers (6–10 workers per nest) and contained immature brood (i.e., eggs, larvae, and pupae).

2.3 | Isolation and Culturing of *B. bassiana* Spores

The biopesticide suspension was prepared by isolating fungal infective conidia from the commercial product Naturalis (Biogard CBC Europe), which contains viable conidiospores of *B. bassiana* strain ATCC 74040 [47], isolated from *Anthonomus grandis* (Boheman), the cotton boll weevil. Petri dishes containing MEA (Malt Extract Agar) as culture medium were plated with 100 μ L of commercial product, which contains at least 2.3×10^7 spores/mL (field recommended rate 8.6×10^5 conidia/mL) and incubated at room temperature ($20^\circ\text{C} \pm 2^\circ\text{C}$) for 3 days. Fungal conidia emerging from plates were collected and re-suspended in sterile water and thus diluted to a known concentration of 10^9 spores/mL, comparable to those used for pest control with different strains of *B. bassiana* [14, 24]. We used unformulated fungal conidia instead of a commercial product to assess the impact of the entomopathogen without the potential effects of formulation adjuvants. We chose to expose individuals a single time to a concentration of fungal spores that falls within the range of those applied in the field for different strains of *B. bassiana*, which can also be higher (10^{10} – 10^{11} conidia/mL) [48–50]. For each experiment, individuals were topically exposed to the fungal biopesticide by applying 1 μ L of spore suspension (exposed: 10^6 spores/ μ L) to the dorsal side of the thorax, while control wasps received 1 μ L of sterile water. Conidia suspension and control solvent were vortexed for 10 s before application to further dissolve potential spore aggregations. Topical exposure was selected over oral exposure to simulate more realistic conditions under which foraging workers might encounter the biopesticide in the field.

2.4 | Recognition Bioassays

2.4.1 | Nestmate Recognition Abilities of Unexposed Colonies Toward Exposed Individuals

To evaluate whether unexposed resident wasps could discriminate between fungus-exposed and unexposed nestmates or foreign individuals, we tested 19 colonies with different lures in presentation bioassays [51, 52]. Workers used as lures were exposed to the conidia suspension (exposed, E, $N=19$ nestmate, $N=19$ foreign) or sterile water (control, C, $N=19$ nestmate, $N=19$ foreign) and then maintained separated from their colonies in a Petri dish with ad libitum water and sugar. One week after the single topical exposure, these workers were killed by freezing (-22°C) to be used as lures in the recognition experiment. We waited for 7 days before sacrificing exposed and unexposed wasps to provide a sufficient time window for the development of possible chemical profile alterations [28]. The head of each freeze-killed wasp was measured under a stereomicroscope (Wild Heerbrugg M8) as a reliable proxy of wasp body size [53], ensuring the selection of lures of comparable size to be presented to each colony. Four different types of lures were presented: a control nestmate, an exposed nestmate, a control foreign individual, and an exposed foreign individual. For each presentation, the lure order was randomized into foil packages numbered 1–4. Presentations were held on a single day from 10 a.m. to 4 p.m. Lures were thawed to room temperature before the trial, and each lure was mounted on forceps and slowly introduced into the nest box approximately 1 cm from the nest [52]. Each lure was used only once. Behavioral responses were

recorded with a camera (Sony DCR-SR35) for the 60 s following the first interaction between any resident wasp on the nest and the presented lure. All behavioral responses of the colony toward lures were analyzed using BORIS software [54] to quantify aggressive behaviors, that is, bites or sting attempts of the lure by colony members.

2.4.2 | Nestmate Recognition Abilities of Colonies Exposed to the Biopesticide

To assess whether resident wasps exposed to the biopesticide *B. bassiana* could distinguish their nestmates from foreign wasps, another presentation experiment was performed on a further 30 nests. Workers were removed from these colonies, exposed to either biopesticide or control solution, reintroduced to their colonies, and then tested with lures representing both unexposed foreign wasps and nestmates. All workers from 15 nests were topically exposed to the fungal suspension, while workers from the other 15 nests received the control solution ($N=107$ workers in “E” nests; $N=106$ in “C” nests). Foundresses were not treated to simulate a more field-realistic situation, as during the postemergence phase, workers leave the nest to forage and may be exposed to the biopesticide, while foundresses remain on the nest to lay eggs and have less chance of direct exposure. For each bioassay, we prepared two lures, a nestmate and a foreign, matched for head size, to assess whether the biopesticide affects the nestmate recognition abilities of exposed workers on the nests. Workers used as lures were removed from their nests before colony exposure. Like the previous experiment, presentations were carried out on a single day from 10 a.m. to 4 p.m., and each presentation was video recorded and analyzed considering the same behaviors.

2.5 | Chemical Analysis

To evaluate the effects of *B. bassiana* on CHCs, a further 42 workers of the same age, from 11 nests used in the first experiment, were exposed either to the biopesticide or a control solution ($N=21$ E and $N=21$ C). After 7 days from the single exposure, they were killed by freezing and their chemical profile was analyzed using Gas Chromatography–Mass Spectrometry (GS-MS). Before analysis, workers were thawed to room temperature, and CHC extracts were obtained by washing each worker in a 2 mL glass vial containing 1 mL of pentane for 15 min [55]. After removing the wasp bodies, the extract was allowed to evaporate. The dried extracts were resuspended in 100 μ L of pentane and transferred to a conical glass insert placed inside the original vial used for extraction. The solvent was then dried under a stream of nitrogen and samples were re-suspended in 20 μ L of a solution containing 40 ng/ μ L of hexadecanol (C_{16}OH) in heptane, which served as the internal standard. One μ L of the extract was injected into a Hewlett-Packard (Palo Alto, CA, USA) 7820A gas chromatograph (GC) coupled to an HP 5977B mass selective detector (using a 70 eV electronic ionization source). A fused ZB-WAX-PLUS (Zebtron) silica capillary column ($60\text{ m} \times 0.25\text{ mm} \times 0.25\text{ mm}$) was installed in the GC. The injector port and transfer line temperatures were set at 200°C and the carrier gas was helium (at 20 PSI head pressure). The temperature protocol was from 50°C to 320°C at a rate of $10^\circ\text{C}/\text{min}$, and the final temperature was held for 5 min. Injections were performed in splitless mode (1 min

purge valve off). Data acquisition and analysis were performed using the Agilent Masshunter Workstation software (version B.07.00) and the open-source software OpenChrom [56], respectively. We used the area of the internal standard as a reference for the relative abundance of each peak and selected only peaks with at least 1% of the peak area of the internal standard. Among the 101 peaks identified, we specifically selected compounds found in at least 25% of the samples [33]. Chromatograms were aligned using the “GCalignR” R package [57], which allowed an initial alignment of the peaks based on their retention time. Peaks corresponding to different compounds in each chromatogram were then manually realigned based on their mass spectra. The mass spectra, along with their corresponding molecular weights, were compared with electronic mass spectra libraries (NIST MS Search v.2.0). Thus, CHCs were identified based on their mass spectra and diagnostic ions.

2.6 | Statistical Analysis

The recorded videos were blind-watched in slow motion (0.5s) using BORIS software [54] and a viewer recorded the total number of aggressive interactions of resident wasps toward lures. For the first experiment, we used the “glmmTMB” package [58] to fit a Generalized Linear Mixed Model (GLMM) using a zero-inflated Poisson distribution. As predictor variables, we used (i) lure treatment (E or C), (ii) nestmateship (foreign or nestmate), (iii) response behaviors, and (iv) caste performing the behavior. For the second experiment, we repeated the same model, replacing lure with worker treatment as the predicted variable. We tested models fit using the “DHARMA” package [59] and “car” package [60] to test which factors of the models had a significant effect on the dependent variable. Then, we carried out post hoc analysis using estimated marginal means with Bonferroni correction, as implemented by the “emmeans” package [61], to examine the statistical differences between the experimental groups. Collected CHCs data were analyzed with a GLMM to evaluate differences in the abundance of single compounds between fungus-exposed and control workers, including colony membership and individual identity as random effects. Multivariate analyses have been applied to the dataset to verify the possibility of attributing fungus-exposed and unexposed specimens to their colony based on the chemical composition. With this aim, we performed a Partial Least Square Discriminant Analysis (PLS-DA), as implemented in the “mixOmics” package [62], to check whether the first principal components were able to capture the differences between groups (fungus-exposed and control specimens) and colony membership based on the chemical composition. All statistical analyses were carried out in R 4.4.0 [63].

3 | Results

3.1 | Nestmate Recognition Toward Exposed Individuals

In the first experiment, four lures were presented to each post-emergence colony: two nestmates and two foreign wasps, with one of each group exposed to the biopesticide and one to the control treatment. In general, exposed lures were attacked more than the control ones (GLMM post hoc, Bonferroni corrected:

$SE=0.0301$, $z=-7.645$, $p<0.0001$), and regardless of the treatment, more aggressive behaviors were recorded toward foreign than nestmate wasps ($SE=0.120$, $z=25.928$, $p<0.0001$) (Figure 1). However, with respect to the biopesticide effects, exposure increased the aggressive behavior toward exposed foreign lures compared to control foreign lures ($SE=0.033$, $z=-4.198$, $p=0.0001$). Treatment also increased the number of aggressive behaviors directed at exposed nestmate lures compared to control nestmates ($SE=0.045$, $z=-6.415$, $p<0.0001$). However, exposed nestmates were attacked less than control foreign lures ($SE=0.117$, $z=13.631$, $p<0.0001$).

3.2 | Nestmate Recognition Abilities of Colonies Exposed to the Biopesticide

In the second experiment, two untreated lures, one nestmate and one foreign, were presented to post-emergence colonies in which all the workers were either exposed to the fungal suspension or the control treatment. Again, colonies responded with aggressive acts toward the presented lures, but the analysis did not show any significant differences in the aggressive response between the two treatments (GLMM post hoc, Bonferroni corrected: aggression toward foreign in control versus exposed nests: $SE=0.258$, $z=0.520$, $p=1$; aggression toward nestmates in control versus exposed nests: $SE=0.199$, $z=-1.626$, $p=0.623$), and the aggressive behavior toward foreign lures resulted higher in both treatments ($SE=1.135$,

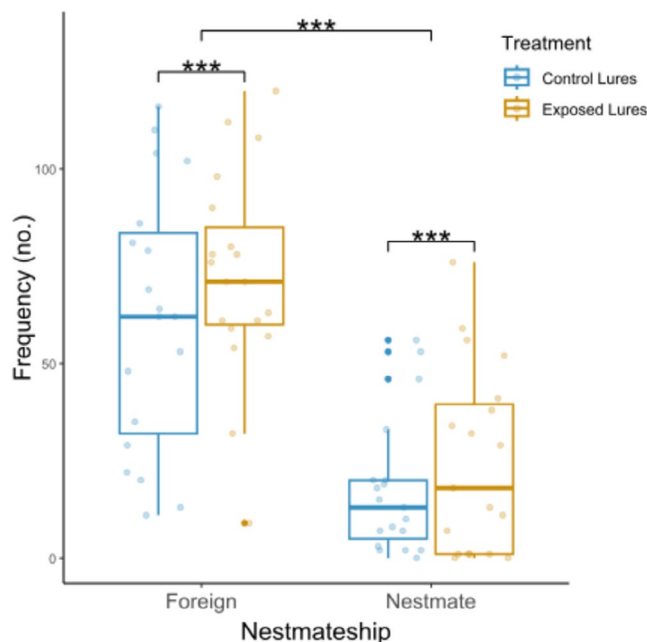


FIGURE 1 | Biopesticide effects on nestmate recognition toward exposed or control individuals. The biopesticide did not affect nestmate recognition but led to an increase in aggressive behaviors toward exposed individuals, both foreign and nestmate. Boxplots show the number of aggressive acts performed by resident wasps toward lures. Thick horizontal lines represent medians, boxes are upper and lower quartiles, and whiskers indicate the highest and lowest values excluding outliers (dark circles). Light dots represent values from single nests scattered to prevent overlap (GLMM post hoc with Bonferroni correction, *** $p<0.0001$).

$z=13.400$, $p<0.0001$) (Figure 2), even when the caste performing the behavior is considered (Supporting Information Section 2.2). Moreover, the ability to discriminate between foreign and nestmate individuals appears to be unaffected in both control and exposed colonies (aggression toward foreign vs. nestmates in control colonies: $SE=2.074$, $z=12.153$, $p<0.0001$; aggression toward foreign vs. nestmates in exposed colonies: $SE=1.178$, $z=7.558$, $p<0.0001$).

3.3 | Chemical Analysis

Statistical analyses were performed on the selected 79 peaks (for the peak selection see paragraph 2.5). The identified compounds belonged to various classes: alkanes ($N=13$), alkenes ($N=4$), methyl-, dimethyl- or trimethyl-branched hydrocarbons ($N=56$), and unidentified compounds ($N=6$) (for more details, see Table S2). No significant differences were found in the relative amount of CHCs between control and exposed workers (GLMM post hoc with Bonferroni correction: $SE=0.137$, $z=1.429$, $p=0.1531$). However, when examining individual compounds, we found that two methyl-alkanes (11-Me-C29: $SE=0.413$, $z=2.533$, $p=0.0112$; 11-Me-C31: $SE=0.586$, $z=2.680$, $p=0.0073$) and one dimethyl-alkane (7, 13-DiMe-C29: $SE=0.615$, $z=2.090$, $p=0.0365$) were significantly lower in the exposed workers than in the control ones. Conversely, one trimethyl-alkane was more abundant in fungus-exposed wasps than in control ones (7, 11, 15-TriMe-C29: $SE=0.149$, $z=-2.395$, $p=0.0165$) (Table 1; for the full analysis and Supporting Information Figures, Section 2.3.2).

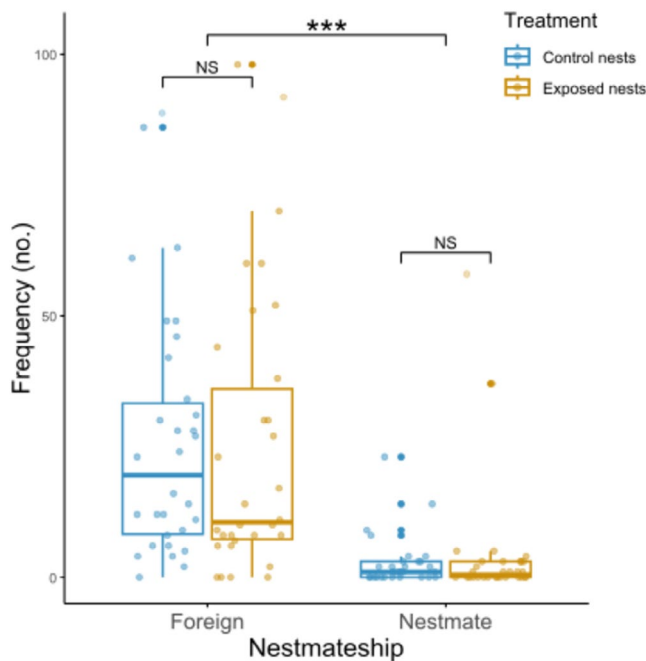


FIGURE 2 | Biopesticide effects on nestmate recognition of exposed or control colonies. Exposure did not affect the aggressive behavior toward foreign and nestmate lures. Boxplots show the number of aggressive acts performed by the colony toward lures. Thick horizontal lines represent medians, boxes are upper and lower quartiles and whiskers indicate the highest and lowest values excluding outliers (dark circles); light dots are values from single nests scattered to prevent overlap (GLMM post hoc with Bonferroni correction, *** $p<0.0001$).

When plotted for the first two principal components, the scores obtained by individuals from different colonies in the PLS-DAs showed that exposed and control individuals were grouped together according to their colonies, showing only marginal distances between groups (Figure 3).

4 | Discussion

Our results show that exposure to the fungal biopesticide *B. bassiana* does not affect the nestmate recognition system in colonies of the predatory paper wasp *P. dominula*. In the first experiment, when exposed or control lures were presented to unexposed colonies, the ability to discriminate between foreign and nestmate lures was maintained regardless of lure exposure. Strikingly, the fungus altered the aggressive response of resident wasps, leading to a uniform increase in attacks toward exposed individuals, either foreign or nestmate. In the second experiment, the overall results indicate no change in the level of discrimination between nestmates and potential intruders, as foreign lures were intensively attacked by both exposed and control colonies. This suggests that resident wasps retain an efficient recognition system. Our GC-MS analyses showed that EF exposure modified the abundance of four methylated hydrocarbons. We hypothesize that these changes in the cuticular profile might convey information about the health status of conspecifics to resident wasps, thereby triggering the increased aggressive behavioral response toward potential pathogen vectors in the nest.

Previous experiments on *Apis mellifera*, *Tetragonisca angustula*, and *Mischocyttarus metathoracicus* have shown how exposure to *B. bassiana* can modify the composition of epicuticular lipids [29, 33, 34]. These findings, alongside our GC-MS results, support the evidence that chemical cues induced by fungus exposure can elicit behavioral responses in our wasps. EF metabolize long-chain hydrocarbons to penetrate the insect cuticle, and such metabolic interactions may be responsible for the observed alterations in CHCs [64]. Nevertheless, unlike what was observed in *A. mellifera*, where exposure to the same fungal biopesticide increased the acceptance of foreign individuals [33], or in *M. metathoracicus*, where wasps could not discriminate between exposed and unexposed conspecifics [29], our exposed lures elicited a higher number of attacks from resident wasps. Thus, our results are consistent with those demonstrated in the stingless bee *T. angustula*, where exposure to *B. bassiana* did not alter the rejection rate of foreign individuals but increased the aggressive response to exposed nestmates [34]. The different responses in our *P. dominula* wasps compared to other species, such as *A. mellifera* honey bees and *Mischocyttarus* paper wasps, may be due to variations in how the fungus affects the chemical profile of exposed individuals between species, as well as the time elapsed since exposure. In honey bees, reduced aggression toward exposed non-nestmates was paired with a decrease of several cuticular alkenes, which are essential for social insect recognition processes [65]. Conversely, *M. metathoracicus* wasps showed no changes in the cuticular profiles of exposed individuals, potentially due to a shorter time interval between exposure and collection for analysis compared to our study (1 and 7 days, respectively). This longer interval was based on previous studies showing increased worker mortality in *P. dominula* [28] and *A. mellifera* [66] after exposure to conidia for more than a week.

TABLE 1 | List of compounds analyzed in *Polistes dominula* workers either exposed to the entomopathogenic fungus *Beauveria bassiana* or unexposed, and their relative abundance (ng/ μ L) in the cuticular mixture.

RT	Compounds	Control, mean $\pm$ SD (ng/ μ L)	Exposed, mean $\pm$ SD (ng/ μ L)	SE	z	p
23.88	C20:1	37.48 $\pm$ 28.81	35.49 $\pm$ 22.37	0.273	0.201	0.84
24.34	Me-C20	5.68 $\pm$ 3.73	4.26 $\pm$ 1.89	0.6	0.723	0.47
26.17	<i>n</i> -C23	2.48 $\pm$ 1.8	2.5 $\pm$ 1.33	0.511	−0.048	0.962
27.81	<i>n</i> -C24	1.99 $\pm$ 1.17	1.69 $\pm$ 0.89	0.662	0.253	0.8
28.83	Me-C24	0.8 $\pm$ 1.31	0.62 $\pm$ 0.8	0.95	0.29	0.771
29.43	<i>n</i> -C25	41.78 $\pm$ 24.37	42.49 $\pm$ 20.7	0.245	−0.06	0.953
29.96	11-Me-C25	8.04 $\pm$ 13.41	5.03 $\pm$ 3.55	0.637	1.051	0.293
30.04	7-Me-C25	1.52 $\pm$ 2.11	1.54 $\pm$ 1.26	0.605	−0.028	0.977
30.18	5-Me-C25	0.96 $\pm$ 1.07	1.08 $\pm$ 0.9	0.624	−0.143	0.886
30.52	3-Me-C25	1.43 $\pm$ 1.51	1.66 $\pm$ 1.22	0.546	−0.172	0.863
30.94	<i>n</i> -C26	18.3 $\pm$ 10.89	19.48 $\pm$ 9.38	0.284	−0.227	0.821
31.43	11-Me-C26	1.14 $\pm$ 1.57	0.69 $\pm$ 0.92	1.106	0.603	0.547
31.87	2-Me-C26	13.67 $\pm$ 17.47	12.17 $\pm$ 7.74	0.363	0.211	0.833
32.21	C27:1	2.18 $\pm$ 2.18	1.71 $\pm$ 2.12	0.78	0.538	0.591
32.48	<i>n</i> -C27	107.94 $\pm$ 46.69	114.01 $\pm$ 51.63	0.192	−0.205	0.837
32.92	11-Me-C27	69.47 $\pm$ 49.28	61.04 $\pm$ 32.87	0.256	0.553	0.58
33.01	7-Me-C27	6.08 $\pm$ 6.97	6.16 $\pm$ 4.52	0.406	−0.052	0.958
33.14	5-Me-C27	11.63 $\pm$ 9.21	9.1 $\pm$ 6.45	0.458	0.677	0.498
33.33	8,11-Di-Me-C27	6.94 $\pm$ 6.82	4.86 $\pm$ 3.39	0.589	0.794	0.427
33.47	3,13-Di-Me-C27	25.22 $\pm$ 13.4	26.33 $\pm$ 15.08	0.277	−0.061	0.951
33.58	5,13-Di-Me-C27	6.59 $\pm$ 8.6	3.27 $\pm$ 3.3	0.859	1.431	0.152
33.84	<i>n</i> -C28	17.69 $\pm$ 7.35	19.15 $\pm$ 10.06	0.288	−0.215	0.83
34.25	13-Me-C28	4.86 $\pm$ 9.25	2.57 $\pm$ 3.14	0.873	1.177	0.239
34.28	11-Me-C28	5.66 $\pm$ 9.45	3.79 $\pm$ 4.93	0.644	0.816	0.415
34.4	7-Me-C28	0.73 $\pm$ 1.35	0.82 $\pm$ 1.25	0.632	−0.212	0.832
34.45	3-Me-C28	1.06 $\pm$ 1.71	0.94 $\pm$ 1.46	0.791	0.189	0.85
34.71	2-Me-C28	24.55 $\pm$ 14.43	28.68 $\pm$ 15.82	0.243	−0.508	0.611
34.85	3,13-Di-Me-C28	1.53 $\pm$ 1.62	2.06 $\pm$ 1.31	0.427	−0.537	0.591
35.1	C29:1	1.33 $\pm$ 2.41	0.81 $\pm$ 1.87	1.003	0.614	0.539
35.22	<i>n</i> -C29	35.82 $\pm$ 19.67	36.13 $\pm$ 21.08	0.272	0.191	0.848
35.65	13-Me-C29	45.47 $\pm$ 53.36	50.45 $\pm$ 57.55	0.232	−0.216	0.829
35.71	11-Me-C29	84.12 $\pm$ 137.57	45.09 $\pm$ 70.06	0.399	2.442	0.015*
35.79	7-Me-C29	34.49 $\pm$ 23.99	31.51 $\pm$ 20.01	0.299	0.446	0.655
35.9	5-Me-C29	12.66 $\pm$ 7.53	11.5 $\pm$ 7.79	0.389	0.367	0.713
36.04	5,11-Di-Me-C29	25.37 $\pm$ 30.81	18.6 $\pm$ 24.38	0.389	0.949	0.343
36.19	7,13-Di-Me-C29	24.16 $\pm$ 24.55	12.3 $\pm$ 21.64	0.613	2.092	0.036*
36.22	7, 11, 15-Tri-Me-C29	9.81 $\pm$ 14.08	24.22 $\pm$ 27.41	0.149	−2.411	0.016*

(Continues)

TABLE 1 | (Continued)

RT	Compounds	Control, mean $\pm$ SD (ng/ μ L)	Exposed, mean $\pm$ SD (ng/ μ L)	SE	z	p
36.3	5,13-Di-Me-C29	13.96 $\pm$ 13.01	10.44 $\pm$ 6.69	0.446	0.737	0.461
36.45	UND2	0.59 $\pm$ 1.02	1.03 $\pm$ 1.85	0.455	-0.655	0.513
36.57	<i>n</i> -C30	8.41 $\pm$ 8.32	7.45 $\pm$ 8.12	0.447	0.415	0.678
36.62	13-Me-C30	6.08 $\pm$ 10.01	3.04 $\pm$ 5.45	0.941	1.519	0.129
36.64	7-Me-C30	4.49 $\pm$ 5.58	7.54 $\pm$ 7.66	0.256	-1.202	0.23
36.92	Me-C30	8.45 $\pm$ 7.38	9.6 $\pm$ 6.89	0.34	-0.237	0.813
36.99	Di-Me-C30	6.25 $\pm$ 9.52	2.96 $\pm$ 4.24	0.939	1.551	0.121
37.33	Et-C30	6.6 $\pm$ 3.3	6.36 $\pm$ 4.87	0.446	0.244	0.807
37.48	3-Me-C30	2.67 $\pm$ 4.49	0.74 $\pm$ 1.76	2.213	1.881	0.06
37.68	C31:1	1.06 $\pm$ 1.56	0.5 $\pm$ 1.14	1.539	0.924	0.355
37.81	<i>n</i> -C31	8.15 $\pm$ 8.31	7.69 $\pm$ 12.92	0.444	0.375	0.708
38.26	13-Me-C31	91.3 $\pm$ 79.5	108.94 $\pm$ 96.92	0.177	-0.75	0.453
38.27	11-Me-C31	37.65 $\pm$ 61.54	20.14 $\pm$ 44.17	0.586	2.691	0.007**
38.36	7-Me-C31	10.79 $\pm$ 10.58	10.99 $\pm$ 11.1	0.362	0.055	0.956
38.47	5-Me-C31	3.04 $\pm$ 4.74	3.31 $\pm$ 7.21	0.508	0.061	0.951
38.54	13,20-Di-Me-C31	14.9 $\pm$ 32.63	18.93 $\pm$ 26.3	0.271	-0.525	0.6
38.63	Di-Me-C31	34.83 $\pm$ 49.93	27.24 $\pm$ 46.57	0.359	1.099	0.272
38.72	7,13-Di-Me-C31	29.07 $\pm$ 27.47	28.73 $\pm$ 23.48	0.286	0.139	0.89
38.96	C33:1	28.93 $\pm$ 21.19	26.08 $\pm$ 18.27	0.316	0.442	0.659
39.12	<i>n</i> -C33	37.48 $\pm$ 28.81	14.53 $\pm$ 8.49	0.342	0.128	0.898
39.42	15-Me-C33	5.68 $\pm$ 3.73	9.53 $\pm$ 6.69	0.345	-0.217	0.828
39.5	7-Me-C33	2.48 $\pm$ 1.8	1.17 $\pm$ 1.85	1.442	1.533	0.125
39.75	Me-C33	1.99 $\pm$ 1.17	2.54 $\pm$ 2.7	0.514	-0.054	0.957
39.92	3-Me-C33	0.8 $\pm$ 1.31	3.38 $\pm$ 4.33	0.731	0.963	0.335
40.64	15-Me-C34	41.78 $\pm$ 24.37	65.19 $\pm$ 64.82	0.224	-0.086	0.932
40.92	13-Me-C34	8.04 $\pm$ 13.41	26.04 $\pm$ 35.41	0.287	-0.033	0.974
40.98	11-Di-Me-C34	1.52 $\pm$ 2.11	19.97 $\pm$ 30.66	0.403	1.212	0.225
41.11	Di-Me-C34	0.96 $\pm$ 1.07	30.58 $\pm$ 32.57	0.318	0.641	0.522
41.22	5,16-Di-Me-C34	1.43 $\pm$ 1.51	19.51 $\pm$ 15.34	0.361	0.644	0.519
41.32	UND3	18.3 $\pm$ 10.89	5.72 $\pm$ 6.48	0.673	1.422	0.155
41.48	<i>n</i> -C35	1.14 $\pm$ 1.57	7.3 $\pm$ 5.07	0.383	-0.102	0.919
41.76	Me-C35	13.67 $\pm$ 17.47	1.72 $\pm$ 2.32	0.636	0.114	0.909
41.85	UND4	2.18 $\pm$ 2.18	1.7 $\pm$ 2.08	1.032	1.156	0.248
42.06	UND5	107.94 $\pm$ 46.69	2.97 $\pm$ 3.04	0.378	-0.636	0.525
42.88	15-Me-C35	69.47 $\pm$ 49.28	12.87 $\pm$ 11.28	0.326	-0.164	0.87
43.18	UND6	6.08 $\pm$ 6.97	33.38 $\pm$ 26.73	0.263	-0.054	0.957
43.32	7-Me-C35	11.63 $\pm$ 9.21	8.52 $\pm$ 9.28	0.59	1.407	0.159
43.43	5-Me-C35	6.94 $\pm$ 6.82	3.26 $\pm$ 3.91	0.681	0.795	0.427

(Continues)

TABLE 1 | (Continued)

RT	Compounds	Control, mean ± SD (ng/μL)	Exposed, mean ± SD (ng/μL)	SE	z	p
43.53	<i>n</i> -C36	25.22 ± 13.4	2.82 ± 4.57	0.673	0.663	0.508
43.7	Me-C36	6.59 ± 8.6	1.1 ± 1.83	1.553	1.531	0.126
45.27	<i>n</i> -C37	17.69 ± 7.35	1.78 ± 2.5	0.77	0.518	0.605
45.62	UND7	4.86 ± 9.25	4.51 ± 4.72	0.226	−1.593	0.111

Note: In bold are the compounds that were statistically different in the two treatments (* $p < 0.05$; ** $p < 0.01$).

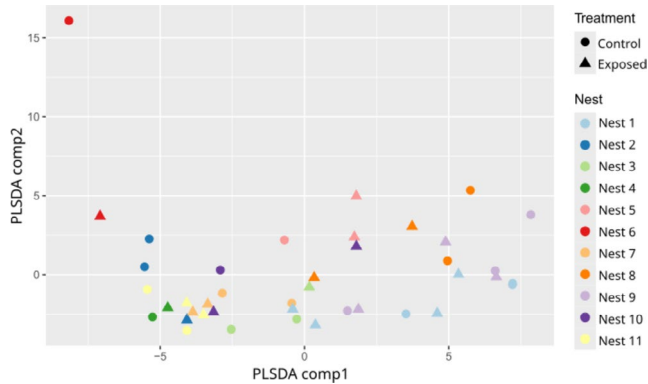


FIGURE 3 | Scatterplot for the first two principal components of a partial least squares discriminant analysis (PLS-DA) based on chemical profiles shows a very weak separation between exposed and control workers of the same nest (for more details, see Supporting Information Section 2.3.1).

and 6 days, respectively. If the health condition of the wasps was indeed impacted by the fungal pathogen, it is plausible that their CHCs profile would also undergo alterations within the same timeframe, as supported by our GC–MS results.

Regarding the nature of the chemical cues, we did not observe a selective alteration of alkenes in the CHCs profile of fungus-exposed workers, as seen in *A. mellifera* [33], nor a reduction of linear alkanes as found in the stingless bee *T. angustula* [34]. Instead, the CHCs changes following exposure consisted of a reduction of several methyl- (11-Me-C29 and 11-Me-C31) or dimethyl-branched (7,13-DiMe-C29) alkanes and the increase of a single compound, the trimethyl-branched alkane 7,11,15-TriMe-C29. It is important to point out that methyl-branched alkanes predominate the CHCs profile of *P. dominula* and play a key role in recognition processes [67]. Even small variations in their proportion can trigger a response from the colony, as demonstrated by previous research showing that minor changes in methyl-branched alkanes elicited aggression toward individuals with altered CHC profiles in this species [37]. The observed reduction in methyl- and dimethyl-branched alkanes, and the significant increase in the trimethyl-branched alkane 7,11,15-TriMe-C29 in our exposed wasps, may therefore have triggered the aggressive response of the resident wasps in our first experiment. Interestingly, 7,11,15-TriMe-C29, the only hydrocarbon that was more abundant in exposed specimens, is not a major component in the CHCs profile of *P. dominula* [68]. Another study showed that artificial wounding in the cuticle of *M. metathoracicus* wasps induces an increase in a single methyl-branched alkane, 3-MeC25, and this subtle change

in the chemical profile was sufficient to change the behavioral response of wasps defending the nest [69]. Similarly, an increase in several methyl-branched CHCs was also recorded in *P. dominula* parasitized by the strepsipteran *Xenos vesparum* [70], and in *A. mellifera* exposed to the parasitic mite *Varroa destructor* [71]. In the latter, the selective increase of the methyl-branched fraction of CHCs induced higher aggressiveness in guard bees toward bee lures with a parasite-altered cuticular profile [71]. Recognition and rejection of parasitized, infected, or wounded individuals appear to be widespread in social insects [69, 71–73], where it represents a crucial first line of defense in social immunity strategies [74, 75]. The stronger aggressiveness shown by our wasps toward exposed lures, both nestmate and foreign, might therefore represent an adaptive response to prevent pathogen outbreaks within the colony.

In our second experiment, when foreign and nestmate lures were presented to exposed or unexposed colonies, we did not observe an altered recognition system. The aggressive behavior toward foreign lures was comparable in both treatments, and exposed colonies retained the ability to reject potential intruders, confirming the unaltered chemical template. Based on our chemical analysis, we can hypothesize that individuals in exposed colonies may have developed an altered CHCs profile without altering or adjusting their nestmate recognition template, thus recognizing nestmate lures as inoffensive and maintaining colony defense and integrity.

Retaining the ability to reject potential intruders does not imply that the biopesticide had no impact on the colony. Previous studies demonstrated that the fungus can indeed affect the overall health status of the colony. Exposure of workers to *B. bassiana* conidia not only resulted in increased mortality of colony individuals but also in a higher rate of larval ejection from nests, suggesting the potential spread of the EF inside the colony [28]. In conclusion, the present study demonstrates how an integrated approach based on behavioral bioassays and chemical analyses can effectively evaluate the sublethal effects of plant protection products. We found that exposure to *B. bassiana* does not disrupt the nestmate recognition system in *P. dominula*, and resident wasps indiscriminately increase aggressive responses toward exposed individuals, regardless of whether they are potential intruders or nestmates. The increased aggressiveness likely represents an adaptive strategy to prevent infected individuals from entering the colony and might be triggered by a small number of altered cuticular compounds that signal the health status of conspecifics to resident wasps. Further research using a holistic approach across various species with different social adaptations and ecological needs is essential to deepen our understanding of the sublethal effects of biopesticides and other contaminants on non-target beneficial organisms. This approach will broaden

our traditional ecotoxicological perspective and enable more comprehensive safety assessments of plant protection products and other potentially harmful substances.

Acknowledgments

We wish to thank Dr. G. Pieraccini, G. Sistri, and F. Calamai for their assistance in the laboratory and sample preparation for chemical analyses. We would like to thank two anonymous Reviewers for their helpful comments on the manuscript.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

1. G. Ceballos, P. R. Ehrlich, A. D. Barnosky, A. García, R. M. Pringle, and T. M. Palmer, “Accelerated Modern Human-Induced Species Losses: Entering the Sixth Mass Extinction,” *Science Advances* 1, no. 5 (2015): e1400253, <https://doi.org/10.1126/sciadv.1400253>.
2. S. T. Turvey and J. J. Crees, “Extinction in the Anthropocene,” *Current Biology* 29, no. 19 (2019): R982–R986, <https://doi.org/10.1016/j.cub.2019.07.040>.
3. C. A. Hallmann, M. Sorg, E. Jongejans, et al., “More Than 75 Percent Decline Over 27 Years in Total Flying Insect Biomass in Protected Areas,” *PLoS One* 12, no. 10 (2017): e0185809, <https://doi.org/10.1371/journal.pone.0185809>.
4. F. Sánchez-Bayo and K. A. G. Wyckhuys, “Worldwide Decline of the Entomofauna: A Review of Its Drivers,” *Biological Conservation* 232 (2019): 8–27, <https://doi.org/10.1016/j.biocon.2019.01.020>.
5. D. L. Wagner, “Insect Declines in the Anthropocene,” *Annual Review of Entomology* 65, no. 1 (2020): 457–480, <https://doi.org/10.1146/annurev-ento-011019-025151>.
6. D. Goulson, “The Insect Apocalypse, and Why It Matters,” *Current Biology* 29, no. 19 (2019): R967–R971, <https://doi.org/10.1016/j.cub.2019.06.069>.
7. T. Glare, J. Caradus, W. Gelernter, et al., “Have Biopesticides Come of Age?,” *Trends in Biotechnology* 30, no. 5 (2012): 250–258, <https://doi.org/10.1016/j.tibtech.2012.01.003>.
8. S. Gupta and A. K. Dikshit, “Biopesticides: An Ecofriendly Approach for Pest Control,” *Journal of Biopesticides* 3 (2010): 186.
9. X. Liu, A. Cao, D. Yan, C. Ouyang, Q. Wang, and Y. Li, “Overview of Mechanisms and Uses of Biopesticides,” *International Journal of Pest Management* 67, no. 1 (2021): 65–72, <https://doi.org/10.1080/09670874.2019.1664789>.
10. D. Tilman, C. Balzer, J. Hill, and B. L. Befort, “Global Food Demand and the Sustainable Intensification of Agriculture,” *Proceedings of the National Academy of Sciences* 108, no. 50 (2011): 20260–20264, <https://doi.org/10.1073/pnas.1116437108>.
11. P. A. Egan, L. V. Dicks, H. M. T. Hokkanen, and J. A. Stenberg, “Delivering Integrated Pest and Pollinator Management (IPPM),” *Trends in Plant Science* 25, no. 6 (2020): 577–589, <https://doi.org/10.1016/j.tplants.2020.01.006>.
12. S. Kumar and A. Singh, “Biopesticides: Present Status and the Future Prospects,” *Journal of Fertilizers & Pesticides* 6 (2015): 129, <https://doi.org/10.4172/jbfbp.1000e129>.
13. S. Kumar, “Biopesticides: A Need for Food and Environmental Safety,” *Journal of Biofertilizers & Biopesticides* 3 (2012): 1–3, <https://doi.org/10.4172/2155-6202.1000e107>.
14. F. Cappa, D. Baracchi, and R. Cervo, “Biopesticides and Insect Pollinators: Detrimental Effects, Outdated Guidelines, and Future Directions,” *Science of the Total Environment* 837 (2022): 155714, <https://doi.org/10.1016/j.scitotenv.2022.155714>.
15. G. Giunti, G. Benelli, V. Palmeri, et al., “Non-Target Effects of Essential Oil-Based Biopesticides for Crop Protection: Impact on Natural Enemies, Pollinators, and Soil Invertebrates,” *Biological Control* 176 (2022): 105071, <https://doi.org/10.1016/j.biocontrol.2022.105071>.
16. L. De Fazi, N. K. Thangasamy, A. Canale, A. Cini, and G. Benelli, “Ecotoxicity of Bioinsecticides to Social Wasps,” *Current Opinion in Environmental Science & Health* 41 (2024): 100575, <https://doi.org/10.1016/j.coesh.2024.100575>.
17. F. Cappa and D. Baracchi, “Bioinsecticides on Honey Bees: Exposure, Sublethal Effects, and Risk Assessment Paradigms,” *Current Opinion in Environmental Science & Health* 41 (2024): 100569, <https://doi.org/10.1016/j.coesh.2024.100569>.
18. H. Gul, S. Saeed, and F. Z. A. Khan, “Entomopathogenic Fungi as Effective Insect Pest Management Tactic: A Review,” *Applied Sciences in Business and Economics* 1 (2014): 10–18.
19. P. A. Shah and J. K. Pell, “Entomopathogenic Fungi as Biological Control Agents,” *Applied Microbiology and Biotechnology* 61, no. 5 (2003): 413–423, <https://doi.org/10.1007/s00253-003-1240-8>.
20. G. M. Mascarin and S. Jaronski, “The Production and Uses of *Beauveria Bassiana* as a Microbial Insecticide,” *World Journal of Microbiology and Biotechnology* 32 (2016): 177, <https://doi.org/10.1007/s11274-016-2131-3>.
21. B. Kendrick, *The Fifth Kingdom* (Hackett Publishing, 2017).
22. “EU Pesticides Database—Active Substances,” European Commission, accessed July 29, 2024, <https://ec.europa.eu/food/plant/pesticides/eu-pesticides-database/start/screen/active-substances>.
23. G. G. Khachatourians, E. P. Valencia, and G. S. Miranpuri, “*Beauveria Bassiana* and Other Entomopathogenic Fungi in the Management of Insect Pests,” in *Microbial Biopesticides*, eds. O. Koul and G. S. Dhaliwal (CRC Press, 2002), 239–275.
24. S. Erler, J. H. Eckert, M. Steinert, and A. T. Alkassab, “Impact of Microorganisms and Entomopathogenic Nematodes Used for Plant Protection on Solitary and Social Bee Pollinators: Host Range, Specificity, Pathogenicity, Toxicity, and Effects of Experimental Parameters,” *Environmental Pollution* 302 (2022): 119051, <https://doi.org/10.1016/j.envpol.2022.119051>.
25. R. E. Brock, A. Cini, and S. Sumner, “Ecosystem Services Provided by Aculeate Wasps,” *Biological Reviews* 96, no. 4 (2021): 1645–1675, <https://doi.org/10.1111/brv.12719>.
26. R. J. Southon, O. A. Fernandes, F. S. Nascimento, and S. Sumner, “Social Wasps Are Effective Biocontrol Agents of Key Lepidopteran Crop Pests,” *Proceedings of the Royal Society B: Biological Sciences* 286, no. 1914 (2019): 20191676, <https://doi.org/10.1098/rspb.2019.1676>.
27. G. V. M. Teixeira, A. R. De Souza, W. F. Barbosa, R. C. Bernardes, and M. A. P. Lima, “Chronic Exposure to a Common Biopesticide Is Detrimental to Individuals and Colonies of the Paper Wasp *Polistes Versicolor*,” *Science of the Total Environment* 810 (2022): 152108, <https://doi.org/10.1016/j.scitotenv.2021.152108>.
28. F. Cappa, L. De Fazi, D. Baracchi, and R. Cervo, “Adverse Effects of the Fungal Biopesticide *Beauveria Bassiana* on a Predatory Social Wasp,” *Science of the Total Environment* 908 (2024): 168202, <https://doi.org/10.1016/j.scitotenv.2023.168202>.
29. A. R. de Souza, A. Prato, W. Franca, et al., “A Predatory Social Wasp Does Not Avoid Nestmates Contaminated With a Fungal Biopesticide,” *Environmental Science and Pollution Research* 30, no. 47 (2023): 103851–103861, <https://doi.org/10.1007/s11356-023-29770-5>.

30. G. J. Blomquist and A. G. Bagnères, *Insect Hydrocarbons: Biology, Biochemistry, and Chemical Ecology* (Cambridge University Press, 2010).
31. J. van Zweden, "Nestmate Recognition in Social Insects and the Role of Hydrocarbons," in *Insect Hydrocarbons: Biology, Biochemistry and Chemical Ecology*, eds. G. J. Blomquist and A. G. Bagnères (Cambridge University Press, 2010), 222–243, <https://doi.org/10.1017/CBO9780511711909.012>.
32. F. Ferrante, E. Pasquini, F. Cappa, L. Bellocchio, and D. Baracchi, "Unravelling the Microplastic Menace: Different Polymers Additively Increase Bee Vulnerability," *Environmental Pollution* 352 (2024): 124087, <https://doi.org/10.1016/j.envpol.2024.124087>.
33. F. Cappa, I. Petrocelli, F. R. Dani, et al., "Natural Biocide Disrupts Nestmate Recognition in Honeybees," *Scientific Reports* 9, no. 1 (2019): 3171, <https://doi.org/10.1038/s41598-019-38963-3>.
34. F. C. R. Almeida, D. M. Magalhães, A. P. Favaris, et al., "Side Effects of a Fungus-Based Biopesticide on Stingless Bee Guarding Behaviour," *Chemosphere* 287 (2022): 132147, <https://doi.org/10.1016/j.chemosphere.2021.132147>.
35. F. Cappa, A. Cini, L. Signorotti, and R. Cervo, "Rethinking Recognition: Social Context in Adult Life Rather Than Early Experience Shapes Recognition in a Social Wasp," *Philosophical Transactions of the Royal Society, B: Biological Sciences* 375, no. 1802 (2020): 20190468, <https://doi.org/10.1098/rstb.2019.0468>.
36. A. Cini, F. Cappa, I. Pepiciello, L. Platania, L. Dapporto, and R. Cervo, "Sight in a Clique, Scent in Society: Plasticity in the Use of Nestmate Recognition Cues Along Colony Development in the Social Wasp *Polistes dominula*," *Frontiers in Ecology and Evolution* 7 (2019): 7, <https://doi.org/10.3389/fevo.2019.00444>.
37. F. R. Dani, G. R. Jones, S. Destri, S. H. Spencer, and S. Turillazzi, "Deciphering the Recognition Signature Within the Cuticular Chemical Profile of Paper Wasps," *Animal Behaviour* 62, no. 1 (2001): 165–171, <https://doi.org/10.1006/anbe.2001.1714>.
38. F. Prezoto, "Management of Social Wasp Colonies in Eucalyptus Plantations (Hymenoptera:Vespidae)," *Sociobiology* 59, no. 4 (2012): 1167–1174, <https://doi.org/10.13102/sociobiology.v59i4.497>.
39. K. B. London and R. L. Jeanne, "The Interaction Between Mode of Colony Founding, Nest Architecture and Ant Defense in Polistine Wasps," *Ethology Ecology and Evolution* 12, no. 1 (2000): 13–25, <https://doi.org/10.1080/03949370.2000.9728440>.
40. J. M. Jandt, E. A. Tibbetts, and A. L. Toth, "Polistes Paper Wasps: A Model Genus for the Study of Social Dominance Hierarchies," *Insectes Sociaux* 61, no. 1 (2014): 11–27, <https://doi.org/10.1007/s00040-013-0328-0>.
41. L. Pardi, "Dominance Order in Polistes Wasps," *Physiological Zoology* 21 (1948): 1–13, <https://doi.org/10.1086/physzool.21.1.30151976>.
42. P. T. Starks and S. Turillazzi, "Polistes Wasps: The Emergence of a Model Genus," *Annales Zoologici Fennici* 43 (2006): 385–386, <https://www.annzool.net/PDF/anz43-free/anz43-385.pdf>.
43. T. R. Armstrong and N. E. Stamp, "Colony Productivity and Foundress Behaviour of a Native Wasp Versus an Invasive Social Wasp," *Ecological Entomology* 28, no. 6 (2003): 635–644.
44. A. Nannoni, R. Cervo, and S. Turillazzi, "Foraging Activity in European Polistes Wasps (Hymenoptera Vespidae)," *Bollettino Della Societa Entomologica Italiana* 133 (2001): 67–78, <https://flore.unifi.it/handle/2158/308909>.
45. A. Urbini, E. Sparvoli, and S. Turillazzi, "Social Paper Wasps as Bio-indicators: A Preliminary Research With Polistes Dominulus (Hymenoptera Vespidae) as a Trace Metal Accumulator," *Chemosphere* 64, no. 5 (2006): 697–703, <https://doi.org/10.1016/j.chemosphere.2005.11.009>.
46. M. C. Lorenzi, M. F. Sledge, P. Laiolo, E. Sturlini, and S. Turillazzi, "Cuticular Hydrocarbon Dynamics in Young Adult *Polistes dominulus* (Hymenoptera: Vespidae) and the Role of Linear Hydrocarbons in Nestmate Recognition Systems," *Journal of Insect Physiology* 50, no. 10 (2004): 935–941, <https://doi.org/10.1016/j.jinsphys.2004.07.005>.
47. F. Mayoral, M. Benuzzi, and E. Ladurner, "Efficacy of the *Beauveria bassiana* Strain ATCC 74040 (Naturalis) Against Whiteflies on Protected Crops," *Bulletin OILB/SROP* 29 (2006): 83–88.
48. S. B. Alves, L. C. Marchini, R. M. Pereira, and L. L. Baumgratz, "Effects of Some Insect Pathogens on the Africanized Honey Bee, *Apis mellifera* L. (Hym., Apidae)," *Journal of Applied Entomology* 120, no. 1–5 (1996): 559–564, <https://doi.org/10.1111/j.1439-0418.1996.tb01652.x>.
49. J. P. Kapongo, L. Shipp, P. Kevan, and B. Broadbent, "Optimal Concentration of *Beauveria bassiana* Vectored by Bumble Bees in Relation to Pest and Bee Mortality in Greenhouse Tomato and Sweet Pepper," *BioControl* 53, no. 5 (2008): 797–812, <https://doi.org/10.1007/s10526-007-9142-9>.
50. J. P. Kapongo, L. Shipp, P. Kevan, and J. C. Sutton, "Co-Vectoring of *Beauveria bassiana* and *Clonostachys rosea* by Bumble Bees (*Bombus impatiens*) for Control of Insect Pests and Suppression of Grey Mould in Greenhouse Tomato and Sweet Pepper," *Biological Control* 46, no. 3 (2008): 508–514, <https://doi.org/10.1016/j.biocontrol.2008.05.008>.
51. F. Cappa, L. Beani, and R. Cervo, "The Importance of Being Yellow: Visual Over Chemical Cues in Gender Recognition in a Social Wasp," *Behavioral Ecology* 27, no. 4 (2016): 1182–1189, <https://doi.org/10.1093/beheco/arw025>.
52. L. Signorotti, F. Cappa, P. d'Ettorre, and R. Cervo, "Novel Insights Into the Ontogeny of Nestmate Recognition in Polistes Social Wasps," *PLoS One* 9, no. 5 (2014): e97024, <https://doi.org/10.1371/journal.pone.0097024>.
53. K. R. Eickwort, "Differential Variation of Males and Females in Polistes Exclamans," *Evolution* 23, no. 3 (1969): 391–405, <https://doi.org/10.2307/2406695>.
54. O. Friard and M. Gamba, "BORIS: A Free, Versatile Open-Source Event-Logging Software for Video/Audio Coding and Live Observations," *Methods in Ecology and Evolution* 7, no. 11 (2016): 1325–1330.
55. F. R. Dani, S. Fratini, and S. Turillazzi, "Behavioural Evidence for the Involvement of Dufour's Gland Secretion in Nestmate Recognition in the Social Wasp *Polistes Dominulus* (Hymenoptera: Vespidae)," *Behavioral Ecology and Sociobiology* 38, no. 5 (1996): 311–319, <https://doi.org/10.1007/s002650050247>.
56. P. Wenig and J. Odermatt, "OpenChrom: A Cross-Platform Open Source Software for the Mass Spectrometric Analysis of Chromatographic Data," *BMC Bioinformatics* 11, no. 1 (2010): 405, <https://doi.org/10.1186/1471-2105-11-405>.
57. M. Ottensmann, M. A. Stoffel, H. J. Nichols, and J. I. Hoffman, "GCalignR: An R Package for Aligning Gas-Chromatography Data for Ecological and Evolutionary Studies," *PLoS One* 13, no. 6 (2018): e0198311, <https://doi.org/10.1371/journal.pone.0198311>.
58. M. E. Brooks, K. Kristensen, K. J. van Benthem, et al., "glmmTMB Balances Speed and Flexibility Among Packages for Zero-Inflated Generalized Linear Mixed Modeling," *R Journal* 9, no. 2 (2017): 378–400.
59. F. Hartig, "DHARMa: Residual Diagnostics for Hierarchical (Multi-Level/Mixed) Regression Models," 2022, <http://florianhartig.github.io/DHARMa/>.
60. J. Fox and S. Weisberg, *An R Companion to Applied Regression*, 3rd ed. (Sage, 2019), <https://socialsciences.mcmaster.ca/jfox/Books/Companion/>.
61. R. V. Lenth, "Emmeans: Estimated Marginal Means, Aka Least-Squares Means," 2022, <https://CRAN.R-project.org/package=emmeans>.
62. F. Rohart, B. Gautier, A. Singh, and K. A. L. Cao, "mixOmics: An R Package for Omics Feature Selection and Multiple Data Integration," *PLoS Computational Biology* 13, no. 11 (2017): e1005752, <https://doi.org/10.1371/journal.pcbi.1005752>.

63. R Development Core Team, *R: A Language and Environment for Statistical Computing* (R Foundation for Statistical Computing, 2008), <http://www.R-project.org>.
64. N. Pedrini, R. Crespo, and M. P. Juárez, “Biochemistry of Insect Epicuticle Degradation by Entomopathogenic Fungi,” *Comparative Biochemistry and Physiology, Part C: Toxicology & Pharmacology* 146, no. 1 (2007): 124–137, <https://doi.org/10.1016/j.cbpc.2006.08.003>.
65. R. Kather and S. J. Martin, “Evolution of Cuticular Hydrocarbons in the Hymenoptera: A Meta-Analysis,” *Journal of Chemical Ecology* 41, no. 10 (2015): 871–883, <https://doi.org/10.1007/s10886-015-0631-5>.
66. D. Carlesso, S. Smargiassi, L. Sassoli, F. Cappa, R. Cervo, and D. Baracchi, “Exposure to a Biopesticide Interferes With Sucrose Responsiveness and Learning in Honey Bees,” *Scientific Reports* 10, no. 1 (2020): 19929, <https://doi.org/10.1038/s41598-020-76852-2>.
67. F. R. Dani, “Cuticular Lipids as Semiochemicals in Paper Wasps and Other Social Insects,” *Annales Zoologici Fennici* 43, no. 5/6 (2006): 500–514.
68. L. Beani, A. G. Bagnères, M. Elia, I. Petrocelli, F. Cappa, and M. C. Lorenzi, “Cuticular Hydrocarbons as Cues of Sex and Health Condition in *Polistes dominula* Wasps,” *Insectes Sociaux* 66, no. 4 (2019): 543–553, <https://doi.org/10.1007/s00040-019-00721-z>.
69. A. R. de Souza, W. França, A. Prato, M. J. Rantala, and F. S. do Nascimento, “Recognition of Conspecific Immunological Status in a Neotropical Paper Wasp,” *Behavioral Ecology* 34, no. 3 (2023): 480–487, <https://doi.org/10.1093/beheco/arad010>.
70. L. Beani, M. M. Lippi, N. Mulinacci, et al., “Altered Feeding Behavior and Immune Competence in Paper Wasps: A Case of Parasite Manipulation?,” *PLoS One* 15, no. 12 (2020): e0242486, <https://doi.org/10.1371/journal.pone.0242486>.
71. F. Cappa, C. Bruschini, I. Protti, S. Turillazzi, and R. Cervo, “Bee Guards Detect Foreign Foragers With Cuticular Chemical Profiles Altered by Phoretic *Varroa* Mites,” *Journal of Apicultural Research* 55, no. 3 (2016): 268–277, <https://doi.org/10.1080/00218839.2016.1229886>.
72. D. Baracchi, A. Fadda, and S. Turillazzi, “Evidence for Antiseptic Behaviour Towards Sick Adult Bees in Honey Bee Colonies,” *Journal of Insect Physiology* 58, no. 12 (2012): 1589–1596, <https://doi.org/10.1016/j.jinsphys.2012.09.014>.
73. M. A. Esparza-Mora, T. Mazumdar, S. Jiang, et al., “Defensive Behavior Is Linked to Altered Surface Chemistry Following Infection in a Termite Society,” *Scientific Reports* 13, no. 1 (2023): 20606, <https://doi.org/10.1038/s41598-023-42947-9>.
74. S. Cremer, C. D. Pull, and M. A. Fürst, “Social Immunity: Emergence and Evolution of Colony-Level Disease Protection,” *Annual Review of Entomology* 63, no. 1 (2018): 105–123, <https://doi.org/10.1146/annurev-ento-020117-043110>.
75. B. Milutinović and T. Schmitt, “Chemical Cues in Disease Recognition and Their Immunomodulatory Role in Insects,” *Current Opinion in Insect Science* 50 (2022): 100884, <https://doi.org/10.1016/j.cois.2022.100884>.

Supporting Information

Additional supporting information can be found online in the Supporting Information section.